

1 Spatial model of Real Estate portfolio exposure - Cholesky decomposition

$$Y(x) = ay(x) + bw(x) \int_0^T Z_w(sx) ds + Z_y(Tx),$$

where $Z_y(tx)$ is a partly time-changed Brownian sheet

$$Z_y(tx) = y \exp(y_1 x_1 + y_2 x_2) W_t^{v_y},$$

with the time changes

$$v_y^2(x_1) = \exp(2y_1 x_1), \quad v_y^3(x_2) = \exp(2y_2 x_2),$$

- deterministic component $ay(x)$, the integral of temperature deviations from the trend $\int_0^T Z_w(sx) ds$ weighed by a factor $bw(x)$, and a noise term $Z_y(Tx)$.

$$\text{Cov}[Z_y(tx)Z_y(sy)] = 2 \exp(y_1(t-s) + y_2(x_2 - y_2)).$$

$x_f = (x_1, \dots, x_d)$ are geographical positions. If the correlation matrix of $Z_y(tx_f)$ has the Cholesky decomposition

$$\text{corr}(Z_y(tx_f), Z_y(tx_k)) = \exp(y_1(x_{f1} - x_{k1}) + y_2(x_{f2} - x_{k2})),$$

then the vector $Z_f y(t)$ has the following representation

$$Z_f y(t) = Z_f y(0) + y W_t^y,$$

where W_t^y is a vector of independent standard Brownian motions. $w \in \mathbb{R}^{p+d}$ is the Cholesky decomposition of the spatial correlation matrix of $Z_w(tx)$ at locations $x_1, \dots, x_p$. Diagonal matrices

$$b_w = \text{diag}(b_w(x_f)), \quad h_w = \text{diag}(w(x_h)), \quad f_w = \text{diag}(w(x_f)),$$

RE thus follows a multivariate normal distribution with moments -

$$E[Y|F_t] = ay + bw \int_0^T Z_f w(s) ds + bw(1 - e^{w(T-t)})Z_f w(t) + Z_f y(t),$$

and

$$\text{Var}[Y|F_t] = \int_0^t (1 - e^{w(T-s)})^2 ds + bw f_w (h_w^T + f_w^T) f_w b_w + 2y(T-t)y^T y.$$

2 Agent Portfolios in Hong Kong

Property Managers - portfolio holdings taken as of 2024 Financial Year, and weighted average extrapolated based on portfolio MV% (market value). The rent valuation time series is proxied by the valuation of nearest estate referenced from the Centaline Big Data transactions database.

The intersection criterion considered is a **ball tree algorithm**, which is a spatial indexing method utilizing **Harviside distance minimization** for spatial proximity. Inputs required are geographical coordinates (latitude and longitude) to create a hyper sphere.

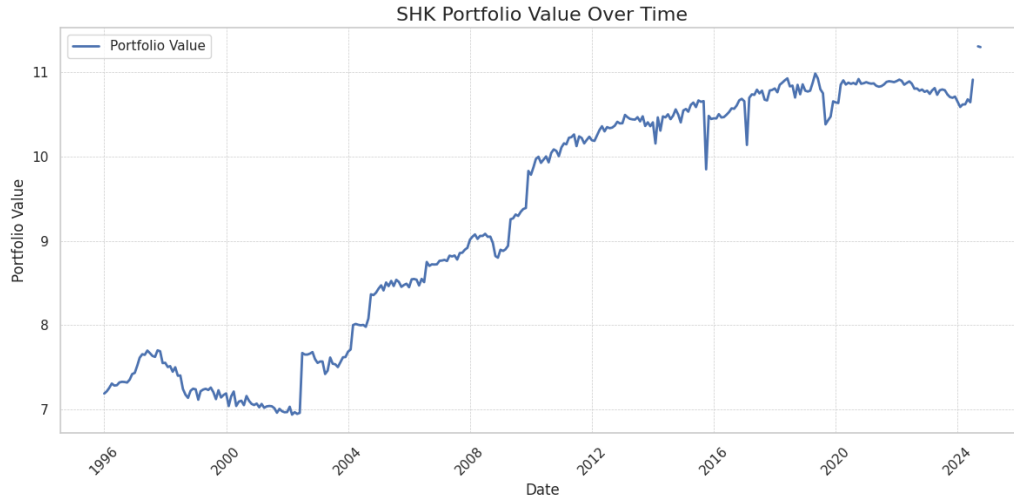


Figure 1: SHK Properties Manager - Time Series of their Hong Kong Portfolio's Rent Valuation. Portfolio as on 2024 FY

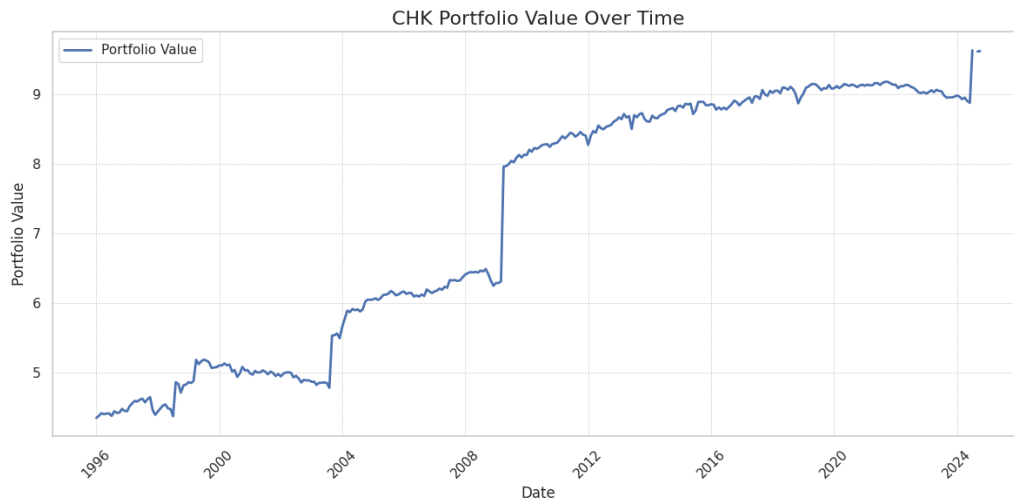


Figure 2: CHK Properties Manager - Time Series of their Hong Kong Portfolio's Rent Valuation. Portfolio as on 2024

Spatial and Temporal Effects Regression - Making the case for Hong Kong's Real Estate's Exposure to Weather

The relationship between housing prices and climatic, temporal, and district characteristics can be expressed as follows:

$$\begin{aligned}\ln(\text{Portfolio rent valuation}_{i,t}) = & \beta_1 \ln(\text{temperature}_{i,t}) + \beta_2 \ln(\text{temperature}_{i,t}) \times \ln(\text{rainfall}_t) + \\ & \beta_3 (\text{number of severe typhoons}_t) \times (\text{number of coastal risk spots}_i) + \\ & \beta_4 \ln(\text{rainfall}_t) \times (\text{number of flooding blackspots}_{i,t}) \times (\text{climate news index}_t) \\ & + \delta(\text{other control variables})_{i,t} + \text{year effect} + \text{district fixed effect} + \epsilon_{i,t}.\end{aligned}$$

District Characteristics (Explaining district fixed effect) The district fixed effect accounts for cross-sectional variations in the following district specific characteristics:

- % built-up
- % vegetated
- % bare soil
- % rock
- % water
- % dense/high vegetation
- % sparse/low vegetation
- Building fraction (%)
- Road/impervious cover fraction (%)
- Average Sky View Factor (SVF)
- Average building height (m)
- Standard deviation of building height (m)

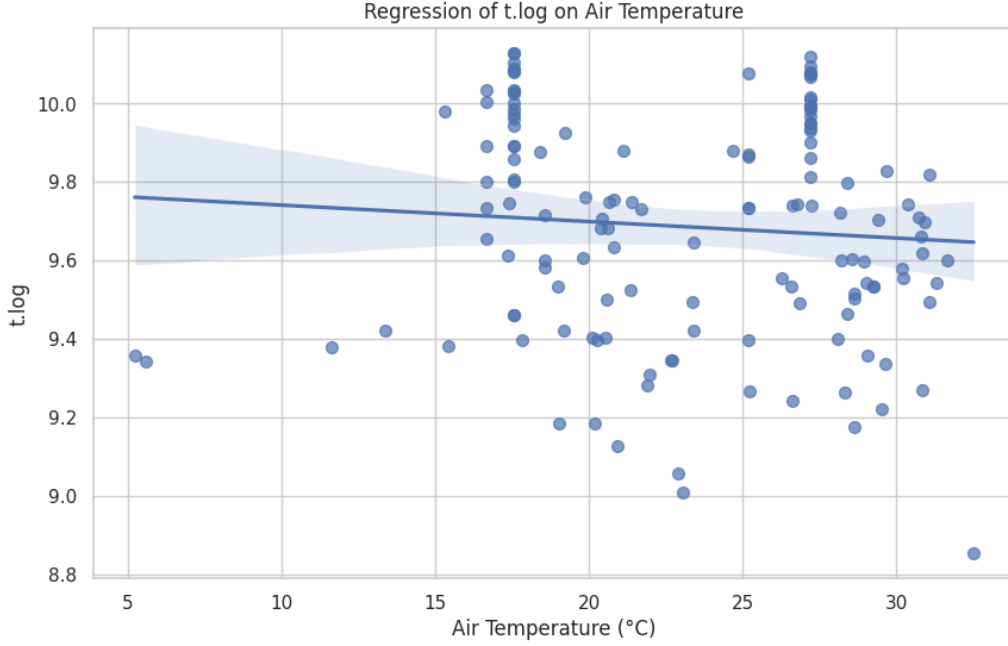


Figure 3: Univariate regression - Basic dependence of Log transformed rent valuation for an estate in **Kennedy Town** Ivy On Belcher' on the bi-hourly temperature time series of Station 6001 (**HK PolyU**). Polynomial method (degree = 2) interpolation used for missing temperature observations.

3 An Empirical Exposure Model

Following W. Hardle (2017) modeled an insurer's income exposure on precipitation from multiple sites, we can assume our agent's income I_1 exhibits a non-linear dependence on the rainfall index $R_{z,i}$, which itself was assumed to follow a Poisson process. Thus, the income model takes the following form:

$$I_1 \sim \log N \{ \mu_I(R_{z,i}), \sigma_I(R_{z,i}) \}$$

where the parameters $\mu_I(R_{z,i})$ and $\sigma_I(R_{z,i})$ are defined as follows:

$$\mu_I(R_{z,i}) = 10 + \prod_{i \in \{1, \dots, n\}} \sum_{z \in \text{Month}} \frac{R_{z,i}}{K_i}$$

$$\sigma_I(R_{z,i}) = 0.1 \left(1 + \sum_{i \in \{1, \dots, n\}} \left(1 - \frac{R_{z,i}}{K_i} \right)^2 \right)$$

Here: - $\mu_I(R_{z,i})$ represents the expected income as a function of the rainfall index $R_{z,i}$, - $\sigma_I(R_{z,i})$ represents the standard deviation of the income, - K_i is the strike price, - $R_{z,i}$ represents the rainfall index for each region i and month z ,

The income of the insurer I_1 follows a log-normal distribution dependent on the underlying rainfall index of multiple regions i

$$R_i \sim \text{Poisson}(\lambda_{Ri})$$

Parameters

Dummy parameters for the model include: - Risk aversion $a_j = a_m = 10^{-4}$, - Strike prices $K_i = 100$ which could be site-specific historical average, - Tick value $h = 1$ monetary units per millimeter of precipitation measured.